

1. System Analysis and Design Concepts

- What is the number of symbols required for a Data Flow Diagram? How is it different from an ERD?
 - How many symbols are used for Data Flow Diagram? Make comparison between DFDs and ERD?
 - How DFD differs from BRD?
 - What are the most common process errors that result from drawing a Data Flow Diagram for a system?
 - Show with example what are the most common process errors occur when a Data Flow Diagram are drawn for a system?
 - Show the cases where illegal data flows happen.
 - Provide examples of data flows that are illegal.
 - What is the necessity of data collection? List several data collection methods.
 - What are the significances of database integrity in System Analysis and Design?
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2. Questionnaire and Interview

- Describe different types of questionnaires. What are the advantages and disadvantages of questionnaires?
 - In system analysis if you use questionnaire as fact-finding process, which advantages and disadvantages you can get?
 - What are the benefits and drawbacks of interviews?
 - What are the advantages and disadvantages of interviews?
 - What should do and avoid for an interview session?
 - What actions should I take and what should I avoid during an interview session?
 - What are the advantages and disadvantages of interviews over other methods?
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3. Time Value of Money / Financial Analysis

- "A dollar today is worth more than a dollar one year from now" — What is the significance of this statement?
 - "A dollar today is worth more than a dollar one year from now" — What is the significance of this Statement? Explain with a suitable example.
 - "The time value of money is not taken into account for Payback Analysis" — Explain the statement with appropriate example.
 - What is Present Value of money? How can we calculate Present Value of money?
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4. Databases vs Files

- Make the distinction between databases and conventional files.

- Differentiate between databases and conventional files.
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5. System Analyst Skills and Tasks

- What are the most important skills that a system analyst must have?
 - Why do many new systems analysts fail to effectively analyse problems? What can they do to become more effective?
 - Show the categories of resources to be allocated to the project.
 - Show the commonly used technique for prioritizing system requirements.
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6. Definitions / Short Notes

Define the following terminologies:

- a) Proxemics
 - b) Brainstorming
 - Body language
 - Spatial zones
 - Office automation systems
 - Expert systems
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7. Reports and Presentations

- Make comparison between factual and administrative format for written report.
 - Give an outline for a standard oral presentation.
 - How reports differ for different persons and places?
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8. Object-Oriented Design and Use Cases

- What is the objective of refining the Use-case model in object design? Why is it important?
 - Describe the steps needed to construct the state chart diagram. Show the relationship between an object state and state transition event.
 - Define the visibility in object-oriented design. Explain the different levels of visibility.
 - Why are the three kinds of objects needed in object-oriented design?
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9. Networking and Frameworks

- What role does network technology play in developing an organization's information system?
- The Capability Maturity Model (CMM) was developed by the Software Engineering Institute at Carnegie Mellon. What is the purpose of the CMM framework and how does it achieve this?

1. Feasibility Analysis

- a) Explain feasibility analysis and the four tests for feasibility analysis with example:
- b) Suppose PSTU wants to impose smartcard to every student for identity and bill/fees paying issues. Discuss the feasibility of this smartcard project as if you are a member of PSTU authority.

2. Report Writing and Presentation

- a) Compare the Factual Format and Administrative Format for written reports.
- b) Write a typical outline and time allocation for an oral presentation.

3. Costs and Benefits

- a) Discuss the types of Costs and Benefits.
- b) Describe the types of cost.
- c) Briefly explain the following techniques:
 - Net benefit analysis
 - Payback analysis
 - Return on Investment
 - Cash flow analysis

4. Questionnaire and Interview

- a) What are the advantages and disadvantages of Questionnaire? What types of questions should be avoided in an interview?
- b) Describe the types of Interview and Interview questions. Discuss the interviewing Do's and Don'ts.
- c) Describe the Do's and Don'ts of interview.

5. Data Flow Diagrams (DFD)

- a) What is DFD? Describe the components of DFD. Differentiate between DFD and Flowchart.
- b) Explain Process Modeling. Differentiate between DFD and Flowchart.
- c) Describe the common process errors on DFD.

6. Break-Even Point and Present Value

- a) What is Break-Even point? Explain the significance of Break-Even point with an example.
- b) What is Break-Even point? Explain the significance of Break-Even point.
- c) \$60,000 benefit two years from now at 15% return from investment means how much benefit is worth today?

- d) \$40,000 benefit two years from now at 15% return from investment means how much benefit is worth today?
- e) Explain Net Present Value.

7. System Requirements

- a) What is system requirement? Describe the types of requirement.

8. Legacy Systems and Information Quality

- a) What is Legacy system?
- b) How to ensure quality of information?

9. Capability Maturity Model (CMM) and Project Manager

- a) Describe Capability Maturity Model. Also explain the levels of maturity.
- b) What are the responsibilities of a project manager?

10. Data, Information, and Knowledge

- a) Distinguish among data, information and knowledge.